Light

Embedded Systems

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OCR GCSE Computer Science — **Embedded Systems (J277/01 Section 1.1.3)**. This topic covers what embedded systems are, their key characteristics, how they use ROM and RAM, why they typically do not use virtual memory, and how the fetch-execute cycle operates within dedicated hardware.

Exam questions typically ask you to **identify** characteristics of embedded systems in a given scenario, **explain** why ROM or RAM is used for specific purposes, and **describe** why certain features (like virtual memory) are or are not suitable. Questions range from 1-mark identifications to 3-mark explanations requiring application to real-world contexts.

What you must know

This checklist covers the key exam-visible concepts. You should be able to explain each point and give an example where appropriate.

- **Define** what an embedded system is and identify examples in everyday devices.
- **State** the key characteristics of embedded systems: single/limited purpose, dedicated hardware, built into a larger system.
- **Explain** why embedded systems use ROM to store firmware and start-up instructions.
- **Explain** why embedded systems use RAM to store data that changes during operation.
- **Describe** how the fetch-execute cycle operates in an embedded system.

- **Explain** why embedded systems typically do not use virtual memory.

> **Specification note.** OCR J277 Section 1.1.3 explicitly requires understanding of embedded systems: their purpose, where they are used, and how they differ from general-purpose computers. The 2024 exam featured an embedded "Follow Me" car system as a major question context.

Toolkit

Embedded system — definition

An **embedded system** is a computer system built into a larger device to perform a **dedicated function**. It has **specific hardware** designed for its task and runs **firmware** stored in ROM. Unlike general-purpose computers, embedded systems cannot easily be reprogrammed by the user.

Key characteristics of embedded systems

Characteristic	Explanation	Example
Single/limited purpose	Designed to perform one specific task or a small set of related tasks	A washing machine controller only controls the wash cycle
Dedicated hardware	Hardware components are chosen specifically for the task; no unnecessary components	A fitness tracker has sensors for heart rate and motion, but no keyboard
Built into a larger system	The computer is a component within a bigger device, not a standalone machine	The CPU in a smart thermostat is inside the thermostat unit
Runs firmware from ROM	Program instructions are stored permanently in ROM and do not change during normal use	A microwave's control program is stored in ROM at manufacture
Limited or no user interface	May have simple buttons/displays or no direct user interaction at all	An anti-lock braking system has no user interface

ROM vs RAM in embedded systems

Memory type	What it stores in an embedded system	Why this type is used
ROM	Firmware, start-up instructions, the main control program	Non-volatile (data retained when power is off); read-only so cannot be accidentally changed
RAM	Currently running data, sensor readings, temporary calculations	Fast read/write access; data can change rapidly during operation

The fetch-execute cycle in embedded systems

Embedded systems use the same fetch-execute cycle as general-purpose computers:

1. **FETCH:** The CPU fetches the next instruction from ROM, where the firmware is stored.
2. **DECODE:** The Control Unit decodes the instruction to determine what operation to perform.
3. **EXECUTE:** The instruction is carried out — this might involve reading data from RAM, performing a calculation, or sending a signal to an actuator.

The cycle repeats continuously while the embedded system is powered on. Instructions always come from ROM (the program), while data being processed is read from and written to RAM.

Virtual memory – why embedded systems don't use it

Virtual memory is a technique where part of secondary storage (like an HDD or SSD) is used as extra RAM when physical RAM is full. Embedded systems typically **do not use virtual memory** because:

- They often have **no secondary storage** (no hard drive or SSD).
- They are designed for a **specific task with known memory requirements** — they should never run out of RAM.
- Virtual memory is **slower** than RAM, which could cause unacceptable delays in real-time systems.
- The system stores **minimal data** at any one time, so RAM is sufficient.

> **Exam tip:** When asked about embedded systems, always link your answer to the specific scenario given. Don't just say "it has dedicated hardware" — say what that hardware is for this device (e.g. "sensors to detect the distance to the car in front").

1. What are embedded systems?

An **embedded system** is a specialised computer system that forms part of a larger device. Unlike a general-purpose computer (like a laptop or desktop), an embedded

system is designed to perform a specific task and cannot easily be reprogrammed by the user.

1.1 Definition and purpose

The word "embedded" means built-in or integrated. An embedded system is a computer that is **built into** another device to control it. The computer is not visible to the user as a separate machine — it is hidden inside the product.

Embedded systems are everywhere in modern life. They control washing machines, microwaves, cars, traffic lights, medical equipment, toys, and smart home devices. Each one contains a processor, memory, and input/output components, but these are tailored specifically for that device's function.

The key difference from a general-purpose computer is that an embedded system has a **single purpose** (or a limited set of related purposes). You cannot install new software on your washing machine or use your car's engine management system to browse the internet.

1.2 Examples of embedded systems

Embedded systems are found in almost every electronic device:

- **Home appliances:** washing machines, dishwashers, microwaves, refrigerators, central heating controllers
- **Vehicles:** engine management systems, anti-lock braking systems (ABS), parking sensors, GPS navigation
- **Consumer electronics:** digital watches, fitness trackers, smart TVs, games consoles
- **Industrial:** factory robots, traffic light controllers, security systems, medical devices

1.3 Key characteristics

Embedded systems share several important characteristics that distinguish them from general-purpose computers:

Single or limited purpose: An embedded system is designed to do one specific job. A traffic light controller only controls traffic lights — it cannot be used to write documents or play games. This makes the system efficient and reliable for its intended task.

Dedicated hardware: The hardware components are chosen specifically for the task. A fitness tracker contains accelerometers and heart rate sensors because it needs them, but it does not have a full keyboard or large display because these are not needed.

Built into a larger system: The embedded computer is a component inside a bigger device. You do not see it as a separate computer — it is integrated into the product. The user interacts with the product, not with the embedded system directly.

Runs firmware: The program that controls an embedded system is called **firmware**. Firmware is stored in ROM and is written at the time of manufacture. It does not change during normal use.

> **Exam tip:** A common 3-mark question asks you to "identify three characteristics of an embedded system" for a given scenario. Make sure each characteristic is different and linked to the device described.

> #### Worked Example 1 – Defining an embedded system
>
> **Question:** State what is meant by an embedded system. [2 marks]

Model answer:
An embedded system is a computer system built into a larger device **(1)** that is designed to perform a specific/dedicated function **(1)**.

Exam tip: For 2-mark definitions, give two distinct points. "Built into a larger device" and "performs a specific function" are the two key ideas the mark scheme expects.

Worked Example 2 — Identifying examples

Question: Give two examples of devices that contain embedded systems. [2 marks]

Model answer:
1. Washing machine **(1)**
2. Car engine management system / ABS / central heating thermostat **(1)**

Exam tip: Do not say "computer" or "laptop" — these are general-purpose computers, not embedded systems. Choose devices where the computer is hidden inside.

Worked Example 3 — Characteristics in context

Question: A smart thermostat controls the heating in a home. It contains an embedded system. Identify two characteristics of an embedded system shown by this device. [2 marks]

Model answer:
1. It has a single/dedicated purpose — it only controls the heating **(1)**.
2. It is built into/part of a larger system — the heating system **(1)**.

Exam tip: Link each characteristic to the specific device in the question. Saying "it has dedicated hardware" without explaining what that means for a thermostat may not score the mark.

Worked Example 4 — Comparing embedded and general-purpose systems

Question: State one difference between an embedded system and a general-purpose computer. [1 mark]

Model answer:
An embedded system is designed for a single/specific purpose, whereas a general-purpose computer can run many different applications **(1)**.

Exam tip: When asked for a difference, make sure you describe both sides. Don't just describe one type of system.

2. Memory in embedded systems

Embedded systems use both ROM and RAM, but for different purposes than a general-purpose computer. Understanding **why** each type of memory is used is a common exam topic.

2.1 ROM in embedded systems

ROM (Read Only Memory) stores the firmware — the permanent program that controls the embedded system. When you turn on a washing machine, the ROM contains the instructions that tell the system how to run each wash cycle.

ROM is used because:

- **Non-volatile:** Data is retained when the power is turned off. The firmware is still there when you next use the device.
- **Read-only:** The program cannot be accidentally deleted or modified during use. This makes the system reliable.
- **Instant availability:** The instructions are ready immediately when the device powers on — no loading time from secondary storage.

The program stored in ROM is written during manufacture and typically does not change. If a manufacturer needs to update the firmware, they may use a special type of ROM that can be rewritten (like EEPROM or flash memory), but this is not something the user does during normal operation.

2.2 RAM in embedded systems

RAM (Random Access Memory) stores data that changes while the embedded system is running. This includes sensor readings, current status information, and temporary values used in calculations.

For example, in a car's "Follow Me" system that maintains distance from the vehicle ahead:

- The **current speed** of the car
- The **distance** to the vehicle in front
- The **speed of the vehicle in front**
- Whether the system is currently **active or inactive**
- Whether the driver has taken **manual control**

All of these values change constantly during operation, so they must be stored in RAM where they can be quickly read and written.

2.3 Why ROM stores the program

In a general-purpose computer, programs are stored on secondary storage (like an SSD) and loaded into RAM when you run them. In an embedded system, this approach would be inefficient:

- Adding secondary storage would increase **cost and size**.
- Loading from storage takes **time**, which is unacceptable for systems that need instant response.
- The program **never changes**, so there is no need for the flexibility of secondary storage. Storing the program directly in ROM means the CPU can begin executing instructions immediately when power is applied. This is essential for devices like car safety systems where any delay could be dangerous.

2.4 The fetch-execute cycle with ROM and RAM

In an embedded system, the fetch-execute cycle works like this:

1. The CPU fetches the next instruction from **ROM** (where the firmware is stored).
2. The Control Unit decodes the instruction.
3. During execution, the CPU may read sensor data from **RAM**, perform calculations, and write results back to **RAM**.
4. The CPU may also send signals to output devices (motors, displays, actuators).

The program always comes from ROM, but the data being processed is in RAM.

> **Real exam context:** The 2024 OCR paper featured a "Follow Me" car system. Questions asked what data would be stored in RAM (speed, distance, sensor readings) and why ROM stores the control program (non-volatile, available immediately, cannot be accidentally changed).

Worked Example 5 — Why ROM stores the program

Question: A fitness tracker contains an embedded system. Explain why ROM is used to store the control program. [2 marks]

Model answer:
ROM is non-volatile so the program is retained when the device is turned off/battery dies **(1)**. ROM is read-only so the program cannot be accidentally deleted or corrupted during use **(1)**.

> **Exam tip:** "Non-volatile" is the key term. Many students lose marks by saying "ROM keeps data when power is off" without using the correct terminology.

Worked Example 6 — What RAM stores

Question: A car has an embedded system that controls automatic braking. The system stores currently running data in RAM. State two examples of data that would be stored in RAM. [2 marks]

Model answer:
1. The current speed of the car **(1)**.
2. The distance to the obstacle/vehicle ahead **(1)**.

Other acceptable answers: whether brakes are currently applied, sensor readings, whether the system is active.

Exam tip: Think about what data **changes** during operation. The control program does not change (that's in ROM), but sensor readings and status values change constantly (that's in RAM).

Worked Example 7 — ROM vs RAM comparison

Question: Explain the difference between how ROM and RAM are used in an embedded system. [2 marks]

Model answer:
ROM stores the firmware/control program/start-up instructions that do not change **(1)**. RAM stores data that changes during operation, such as sensor readings or current status **(1)**.

Exam tip: The key distinction is: ROM = program (permanent), RAM = data (temporary/changing). Always give an example of what each stores.

Worked Example 8 — Start-up in an embedded system

Question: Describe what happens when an embedded system is first powered on. [3 marks]

Model answer:
The CPU fetches the first instruction from ROM **(1)**. The firmware/start-up program stored in ROM begins executing **(1)**. The system initialises hardware components and begins its dedicated task **(1)**.

Exam tip: Remember that embedded systems do not "boot" like a PC loading Windows. The program is already in ROM and begins executing immediately.

3. Applications, virtual memory, and analysis

Higher-mark questions require you to **apply** your knowledge of embedded systems to specific scenarios and **explain** why certain design choices are made.

3.1 Why embedded systems don't use virtual memory

Virtual memory is a technique used in general-purpose computers where part of secondary storage acts as extra RAM. When RAM is full, data is swapped to the hard

drive/SSD.

Embedded systems typically do **not** use virtual memory for several reasons:

No secondary storage: Most embedded systems do not have a hard drive or SSD. There is nowhere to create virtual memory. A washing machine controller has ROM and RAM, but no disk storage.

Known memory requirements: Embedded systems are designed for a specific task with predictable data needs. The designers know exactly how much RAM is required and include enough. The system should never "run out" of RAM.

Speed requirements: Virtual memory is much slower than RAM because it involves reading from/writing to secondary storage. For real-time systems like car brakes or medical monitors, this delay could be dangerous.

Minimal data storage: Embedded systems only store the data they need for their current task. They do not run multiple applications simultaneously like a general-purpose computer.

3.2 Dedicated hardware

The hardware in an embedded system is specifically chosen for its task. This is more efficient than general-purpose hardware because:

- **Lower cost:** No unnecessary components are included.
- **Lower power consumption:** Important for battery-powered devices like fitness trackers.
- **Smaller size:** The system can fit inside the host device.
- **Optimised performance:** Hardware is matched to the task requirements.

3.3 Real-world scenario: Follow Me car system

The 2024 OCR exam featured an embedded system in a car that automatically follows the vehicle ahead at a safe distance. Analysing this example:

Characteristics shown:

- **Single purpose:** The system only maintains safe distance — it does not play music or navigate.
- **Dedicated hardware:** Distance sensors, speed sensors, and actuators for braking/acceleration are specific to this task.
- **Built into a larger system:** The embedded computer is inside the car, not a separate device.

Memory usage:

- **ROM:** Stores the control algorithm that decides when to speed up or slow down.
- **RAM:** Stores current speed, distance to vehicle ahead, speed of vehicle ahead, whether driver has control.

Why no virtual memory:

- The car's embedded system has no secondary storage to use as virtual memory.
- The system stores very few data items at any time — RAM is sufficient.
- Real-time response is critical — any delay from accessing virtual memory could cause an accident.

> **Exam tip:** The 2024 examiner's report noted that some candidates thought virtual memory could not be used because there was "no internet connection". This is incorrect — virtual memory has nothing to do with the internet. It uses secondary storage, not network access.

Worked Example 9 — Why no virtual memory

Question: An embedded system in a medical heart monitor does not use virtual memory. Explain why virtual memory would not be suitable for this device. [3 marks]

Model answer:
The embedded system is unlikely to have secondary storage, so there is nowhere to create virtual memory **(1)**. The system stores minimal data (just current heart rate readings), so it will not run out of RAM **(1)**. Virtual memory is slower than RAM, and any delay in a medical monitor could be dangerous **(1)**.

Exam tip: For "explain why not" questions, give reasons specific to embedded systems, not general statements. "It's too slow" alone is not enough — explain why speed matters for this device.

Worked Example 10 — Dedicated hardware

Question: A smart doorbell contains an embedded system with dedicated hardware. Explain what is meant by dedicated hardware and why it is used. [3 marks]

Model answer:
Dedicated hardware means the components are specifically designed/chosen for the doorbell's task **(1)**. This includes a camera, microphone, speaker, and motion sensor — only what is needed **(1)**. Using dedicated hardware reduces cost/size/power consumption compared to using general-purpose components **(1)**.

Exam tip: When asked about dedicated hardware, name specific components that would be in this device. This shows you understand the concept and can apply it.

Worked Example 11 — Applying characteristics to a scenario

Question: An automatic garden watering system contains an embedded system. It uses sensors to detect soil moisture and activates a pump when the soil is dry. Identify three characteristics of embedded systems and explain how each applies to this device. [6 marks]

Model answer:
1. Single/limited purpose: The system only controls garden watering — it detects moisture and activates the pump. It cannot perform other tasks like

browsing the internet **(2)**.
2. Dedicated hardware: The system contains moisture sensors and pump controllers specifically needed for this task. It does not include unnecessary components like a keyboard or monitor **(2)**.
3. Built into a larger system: The embedded computer is a component inside the watering system. The user interacts with the watering system, not directly with the embedded processor **(2)**.

Exam tip: For 6-mark questions, you need to identify the characteristic AND explain how it applies to the specific scenario. Each characteristic with its application is worth 2 marks.

Worked Example 12 — Extended analysis (6-mark style)

Question: A factory uses an embedded system to control a robotic arm on a production line. The robot arm picks up components and places them in the correct position. Explain how the embedded system uses ROM and RAM, and why it does not use virtual memory. [6 marks]

Plan:

- Explain what ROM stores and why
- Explain what RAM stores and why
- Explain why virtual memory is not used

Model answer:
The embedded system uses ROM to store the firmware — the control program that tells the robot arm how to move and where to place components. ROM is used because it is non-volatile, meaning the program is retained when the system is powered off, and because it is read-only, preventing accidental changes to the control software.
RAM is used to store data that changes during operation. This includes the current position of the arm, sensor readings about component locations, and status information about whether a component has been picked up or placed.
Virtual memory is not suitable for this system because the embedded controller has no secondary storage where virtual memory could be created. Additionally, the system has predictable memory requirements — the designers have included sufficient RAM for the known data needs. Virtual memory would also introduce delays that could disrupt the precise timing required for robotic movements.

Exam tip: Extended questions require full explanations, not just lists. Write in paragraphs and explain the reasoning behind each point. The examiner's

report emphasises that candidates who give reasons score higher than those who only state facts.

Worked examples summary

The 12 worked examples above cover the main question types you will encounter on embedded systems. Here is a summary of the key question patterns:

Example	Question type	Key skill
1	Define (2 marks)	Two key points: built-in + dedicated function
2	Give examples (2 marks)	Name devices with hidden computers, not PCs
3	Identify characteristics (2 marks)	Link characteristics to the scenario
4	State difference (1 mark)	Compare both systems, not just one
5	Explain ROM usage (2 marks)	Non-volatile + read-only
6	State RAM data (2 marks)	Data that changes during operation
7	ROM vs RAM (2 marks)	Program vs changing data
8	Describe start-up (3 marks)	Fetch from ROM, immediate execution
9	Explain no virtual memory (3 marks)	No storage, minimal data, speed critical
10	Dedicated hardware (3 marks)	Define + give examples + benefits
11	Apply characteristics (6 marks)	Three characteristics with context
12	Extended analysis (6 marks)	ROM + RAM + virtual memory reasoning

Quick reference

Key terms — exam-ready definitions

Term	Definition (mark scheme style)
Embedded system	A computer system built into a larger device to perform a dedicated/specific function
Firmware	Permanent software stored in ROM that controls an embedded system; written at manufacture
Dedicated hardware	Hardware components specifically designed/chosen for a particular task

Term	Definition (mark scheme style)
ROM (in embedded systems)	Stores the firmware/control program; non-volatile; read-only during normal operation
RAM (in embedded systems)	Stores currently running data such as sensor readings and status values; volatile; read/write
Non-volatile	Memory that retains its contents when power is removed
Volatile	Memory that loses its contents when power is removed
Virtual memory	A technique where secondary storage is used as extra RAM when physical RAM is full
Single purpose	Designed to perform one specific task or a limited set of related tasks
General-purpose computer	A computer that can run many different applications and be used for various tasks
Fetch-execute cycle	The continuous process where the CPU fetches, decodes, and executes instructions
Sensor	Input device that detects physical conditions (temperature, motion, distance, etc.)
Actuator	Output device that causes physical movement or action (motor, valve, etc.)

Embedded system examples by category

Category	Examples	What ROM stores	What RAM stores
Home appliances	Washing machine, microwave, thermostat	Control programs for cycles/temperatures	Current settings, timer values, temperatures
Vehicle systems	ABS, engine management, parking sensors	Safety algorithms, control logic	Speed, distance, sensor readings
Consumer devices	Fitness tracker, smart watch, digital camera	Operating firmware, display routines	Step count, heart rate, current image
Industrial	Traffic lights, factory robots, security systems	Timing sequences, movement patterns	Current state, sensor inputs, position data

Why embedded systems do not use virtual memory — summary

Reason	Explanation
No secondary storage	Most embedded systems have no HDD/SSD where virtual memory could be created
Known memory requirements	Designers include exactly the RAM needed; the system should never run out
Speed requirements	Virtual memory is slower than RAM; delays are unacceptable in real-time systems
Minimal data	Only stores data for current task; does not run multiple applications

Typical pitfalls

Pitfall 1 — Confusing embedded systems with general-purpose computers.
Wrong: "A laptop is an embedded system because it is portable."
Right: A laptop is a **general-purpose computer** that can run many applications. An embedded system performs a **single dedicated function** and is built into another device.

Pitfall 2 — Saying ROM stores "data".
Wrong: "ROM stores data in an embedded system."
Right: ROM stores the **firmware/program/instructions**. RAM stores the data that changes during operation.

Pitfall 3 — Thinking virtual memory needs internet.
Wrong: "Embedded systems don't use virtual memory because they have no internet connection."
Right: Virtual memory uses **secondary storage**, not the internet. Embedded systems don't use it because they typically have **no secondary storage**.

Pitfall 4 — Not linking characteristics to the scenario.
Wrong: "It has dedicated hardware." (too vague)
Right: "It has dedicated hardware such as **distance sensors and brake actuators** specifically needed for maintaining safe distance."

Pitfall 5 — Forgetting to explain WHY ROM is used.
Wrong: "ROM stores the program."
Right: "ROM stores the program because it is **non-volatile** (retained when power is off) and **read-only** (cannot be accidentally changed)."

Pitfall 6 — Giving generic examples of RAM data.
Wrong: "RAM stores information."
Right: For a car braking system, RAM stores **current speed**, **distance to obstacle**, and **brake status**. Be specific to the scenario.

Pitfall 7 — Thinking embedded systems can be reprogrammed by users.
Wrong: "Users can install new apps on an embedded system."
Right: Embedded systems run **firmware set at manufacture**. Users cannot install new software — the system performs its dedicated function only.

Pitfall 8 — Confusing "single purpose" with "limited capability".
Wrong: "Embedded systems are single purpose because they are not powerful."
Right: "Single purpose" means designed for **one specific task**. An embedded system in a car may be very powerful but is still single purpose.

Pitfall 9 — Not explaining both sides when asked for a difference.
Wrong: "Embedded systems are built into devices."
Right: "Embedded systems are built into devices, **whereas** general-purpose computers are standalone machines that can be used for many tasks."

Pitfall 10 — Saying "faster" without explaining what is faster.
Wrong: "ROM is used because it's faster."
Right: ROM provides **instant access to the program** on power-up — no loading time from secondary storage is needed.

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